

QUIZ SOLUTIONS

Quiz 1

Q1. In machine learning, what term refers to a situation where a model learns the training data too well, capturing noise and leading to poor generalization on new data?

Correct Answer: (d) Overfitting

Q2. What is the purpose of name mangling in Python classes?

Correct Answer: (d) Preventing access to class attributes

Q3. What is the primary purpose of using nested loops in programming?

Correct Answer: (c) To execute a loop inside another loop

Q4. In Python, what is the purpose of file handling?

Correct Answer: (d) To read from and write to files on disk

Q5. When integrating Azure AI services into a CI/CD pipeline, which practice involves automating the deployment of AI models and services?

Correct Answer: (a) CI/CD Integration

Q6. In NLP, what is the process of reducing words to their base or dictionary form while ensuring they are still valid words?

Correct Answer: (a) Lemmatization

Q7. In statistics, which term refers to the function that describes the likelihood of various outcomes in a random experiment?

Correct Answer: (a) Probability distribution

Q8. Which measure provides an indication of how spread out the values in a dataset are?

Correct Answer: (d) Standard Deviation

Q9. What is the primary purpose of string operations in programming?

Correct Answer: (d) To manipulate and process text

Q10. What is the process of breaking a text into individual words or tokens?

Correct Answer: (c) Word tokenization

Q11. Which command is used to display the contents of a directory in the terminal?

Correct Answer: (a) ls

Q12. In univariate linear regression with vectorization, what advantage does vectorization offer over loops?

Correct Answer: (c) Improved computational efficiency

Q13. Which deep learning framework is developed by Google?

Correct Answer: (c) TensorFlow

Q14. Which service can you use to analyze images and extract text from them in Microsoft Azure?

Correct Answer: (d) Content Moderator

Q15. In Pandas, which type of join combines rows from two DataFrames based on common columns, keeping only matching rows?

Correct Answer: (c) Inner join

Q16. What type of solution can you create using Microsoft Azure's Conversational AI services?

Correct Answer: (a) Speech recognition

Q17. Which Azure service provides text-to-speech, translation, and focus mode features?

Correct Answer: (d) Azure Immersive Reader

Q18. In OOP, what is an abstract method?

Correct Answer: (c) A method that has no implementation

Q19. Which measures assess the spread or dispersion of data points around the mean?

Correct Answer: (c) Variance and Standard Deviation

Q20. Which measure indicates how many standard deviations a data point is away from the mean?

Correct Answer: (b) Z-Score

Blank Quiz 1

Q1. What does broadcasting in Numpy allow you to do?

Correct Answer: (b) Perform element-wise operations on arrays with different shapes

Q2. In Linux, which command is used to update the package list?

Correct Answer: (a) apt-get

Q3. Which algorithm is used for both classification and regression and finds the best hyperplane?

Correct Answer: (b) Support Vector Machine (SVM)

Q4. What is the main goal of machine learning?

Correct Answer: (b) To enable computers to learn from data and improve performance over time

Q5. Which Numpy function is used to create a copy of an array?

Correct Answer: (b) `numpy.copy()`

Q6. What is an inner/nested class in OOP?

Correct Answer: (a) A class defined within another class

Q7. What is the average value of a dataset called?

Correct Answer: (c) Mean

Q8. What method is used to select elements from a Numpy array based on a condition?

Correct Answer: (d) `where()`

Q9. What is chunking in NLP?

Correct Answer: (a) Chunking

Q10. Which term describes a variable with distinct separate values?

Correct Answer: (c) Discrete

Q11. Which library is commonly used for NLP tasks?

Correct Answer: (d) NLTK

Q12. Which Seaborn function creates a bar plot?

Correct Answer: (b) sns.barplot()

Q13. Which preprocessing step converts all characters to lowercase?

Correct Answer: (d) Lowercasing

Q14. Which mechanism focuses on important parts of input sequences?

Correct Answer: (b) Attention Mechanism

Q15. Which functions introduce nonlinearity in neural networks?

Correct Answer: (b) Activation functions

Q16. Which field focuses on understanding and generating human language?

Correct Answer: (c) Natural Language Processing

Q17. Which Azure service provides language translation?

Correct Answer: (d) Azure Translator

Q18. Which data structure is commonly used in Pandas to store tabular data?

Correct Answer: (c) DataFrames

Q19. Which measure quantifies relative variability?

Correct Answer: (a) Coefficient of Variation

Q20. Which graphical representation visualizes mathematical operations in neural networks?

Correct Answer: (d) Computational Graph

Quiz 2

Q1. What serves as the entry point for accessing cloud service features?

Correct Answer: (b) Default Endpoint

Q2. Which command changes permissions of a file?

Correct Answer: (c) chmod

Q3. Which NLP model predicts a target word from surrounding context?

Correct Answer: (a) Continuous Bag-of-Words (CBOW)

Q4. What is dynamic polymorphism?

Correct Answer: (b) The ability of a subclass to override methods of its base class

Q5. What is structured data?

Correct Answer: (d) Data organized into a specific format or schema

Q6. Which cloud service is suitable for real-time object detection?

Correct Answer: (c) Azure Cognitive Services

Q7. In univariate regression without vectorization, how is gradient calculated?

Correct Answer: (c) Using a loop to iterate over each data point

Q8. Which Seaborn function creates a heatmap?

Correct Answer: (d) sns.heatmap()

Q9. Which regex symbol matches any single character?

Correct Answer: (c) .

Q10. What are universal array functions in Numpy?

Correct Answer: (b) Functions that operate element-wise on arrays

Q11. Which Azure service deploys ML models as REST APIs?

Correct Answer: (d) Azure Machine Learning Service

Q12. Which field studies human language understanding?

Correct Answer: (c) Natural Language Processing

Q13. Which technology packages applications with dependencies?

Correct Answer: (a) Docker Containers

Q14. Which technique handles missing data in Pandas?

Correct Answer: (b) Data imputation

Q15. Which NLP technique converts words into dense vectors?

Correct Answer: (a) Word2Vec

Q16. What is the main goal of exception handling?

Correct Answer: (b) To gracefully handle errors and exceptions

Q17. What is the process of identifying named entities?

Correct Answer: (a) NER tagging

Q18. Which Pandas operation performs element-wise addition?

Correct Answer: (b) df.add()

Q19. What is multilevel inheritance?

Correct Answer: (b) Inheriting from a child class

Q20. Which Pandas method saves a DataFrame to CSV?

Correct Answer: (c) df.to_csv()

Quiz 4

Q1. What is the main application of NLP solutions?

Correct Answer: (c) Text analysis and understanding

Q2. What is feature selection?

Correct Answer: (b) Feature selection

Q3. Which measure of central tendency is most affected by outliers?

Correct Answer: (b) Mean

Q4. Which Azure service filters inappropriate content?

Correct Answer: (b) Azure Content Moderator

Q5. Which NLP model predicts a target word from context?

Correct Answer: (a) Continuous Bag-of-Words (CBOW)

Q6. What is the advantage of functions?

Correct Answer: (d) They allow you to reuse code and improve code organization

Q7. What lexical database provides semantic word relationships?

Correct Answer: (d) WordNet

Q8. What is the primary purpose of Numpy?

Correct Answer: (b) To perform mathematical operations on arrays and matrices

Q9. What problem occurs when gradients become very small?

Correct Answer: (b) Vanishing gradients

Q10. Which RNN processes input in forward and backward directions?

Correct Answer: (a) Bi-Directional LSTM/RNN

Q11. How does Azure Anomaly Detector work?

Correct Answer: (d) It uses machine learning to predict anomalies

Q12. Which Azure service provides sentiment analysis and NER?

Correct Answer: (c) Azure Text Analytics

Q13. What process converts continuous data into intervals?

Correct Answer: (a) Binning

Q14. Which type of data contains labels or categories?

Correct Answer: (c) Categorical data

Q15. Which command displays file contents in Unix?

Correct Answer: (c) cat

Q16. What is a function defined inside another function called?

Correct Answer: (a) Nested function

Q17. Which Numpy function performs element-wise addition?

Correct Answer: (a) numpy.add()

Q18. Which loss function is common for classification tasks?

Correct Answer: (d) Cross-Entropy

Q19. Which sequence model is commonly used in machine translation?

Correct Answer: (b) Recurrent Neural Networks (RNN)

Q20. Which command changes file permissions?

Correct Answer: (c) chmod

Blank Quiz 2

Q1. What is the primary purpose of AI?

Correct Answer: (d) To perform tasks without any human intervention

Q2. Which approach manages authentication for cloud resources?

Correct Answer: (a) IAM Policies

Q3. Which keyword introduces an alternative condition in Python?

Correct Answer: (c) Else

Q4. What is the role of gradient descent?

Correct Answer: (c) To minimize the cost function and find optimal parameters

Q5. What are the two main steps in neural network training?

Correct Answer: (b) Forward and Backward Passes

Q6. Which Python library offers a high-level API for neural networks?

Correct Answer: (a) Keras

Q7. Which measure quantifies linear relationship strength?

Correct Answer: (b) Correlation Coefficient

Q8. What is the purpose of list comprehension?

Correct Answer: (b) To manipulate and filter lists using a concise syntax

Q9. Which linguistic component governs sentence structure?

Correct Answer: (c) Syntax

Q10. What is the purpose of a function?

Correct Answer: (a) To organize code into reusable blocks

Q11. Which type of data represents quantities?

Correct Answer: (d) Quantitative data

Q12. Which measure shows relative position in a dataset?

Correct Answer: (c) Percentile

Q13. Which probability considers events occurring together?

Correct Answer: (a) Joint probability

Q14. What is the primary task in supervised learning?

Correct Answer: (d) To learn from labeled training data to make predictions

Q15. Which plot shows distribution of a single variable?

Correct Answer: (d) Univariate plot

Q16. Which cloud service supports sentiment analysis, NER, and translation?

Correct Answer: (b) Azure Cognitive Services

Q17. What is hierarchical inheritance?

Correct Answer: (a) Inheriting from a single parent class

Q18. Which Responsible AI practice ensures transparency?

Correct Answer: (b) Explainability and Transparency

Q19. Which decision boundary is learned in logistic regression?

Correct Answer: (b) Linear decision boundary

Q20. Which loss function is used for classification?

Correct Answer: (d) Cross-Entropy

Quiz 6

Q1. Which Python function applies a function to all iterable elements?

Correct Answer: (d) map()

Q2. Which lexical database provides semantic relationships?

Correct Answer: (d) WordNet

Q3. Which neural network is commonly used for image classification?

Correct Answer: (a) 2D Convolutional Neural Network (CNN)

Q4. Which Pandas operation groups data and applies aggregate functions?

Correct Answer: (a) GroupBy

Q5. Which Azure service manages identity and access?

Correct Answer: (d) Azure Active Directory

Q6. What is the purpose of dunder methods?

Correct Answer: (d) Overloading operators and providing special behavior

Q7. Which Azure service builds and deploys ML models?

Correct Answer: (a) Azure Machine Learning

Q8. Which ML problem predicts continuous numerical output?

Correct Answer: (a) Regression

Q9. Which Azure service supports speech-to-text and speaker recognition?

Correct Answer: (c) Azure Speech Service

Q10. Which interface provides a GUI for Azure resource creation?

Correct Answer: (b) Azure Portal

Q11. What is a function inside another function called?

Correct Answer: (a) Nested function

Q12. What is an object in OOP?

Correct Answer: (c) An instance of a class with its own properties and methods

Q13. What is reducing words to root form called?

Correct Answer: (b) Stemming

Q14. Which Seaborn function creates a histogram with KDE?

Correct Answer: (b) sns.histplot()

Q15. What are class variables?

Correct Answer: (a) Variables that are shared among all instances of a class

Q16. Why select the appropriate Azure AI service?

Correct Answer: (b) To ensure the responsible use of AI technologies

Q17. What is dividing text into words in languages without spaces called?

Correct Answer: (d) Word segmentation

Q18. Which Pandas structure stores one-dimensional labeled data?

Correct Answer: (d) Series

Q19. Seaborn is based on which library?

Correct Answer: (b) Matplotlib

Q20. Which plot displays distribution for a single variable?

Correct Answer: (d) Univariate plot

LAB SOLUTIONS

Lab 01

Q1. GRU-Based Named Entity Recognition Model

```
# Import required libraries
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, GRU, Dense
import numpy as np

# Sample sentences
sentences = [
    "John lives in New York",
    "Google is located in California"
]

# Named Entity labels
# 1 = Person, 2 = Place, 3 = Organization, 0 = Other
labels = [
    [1, 0, 0, 2],
    [3, 0, 0, 2]
]

# Tokenize words
word_index = {}
index = 1

for sentence in sentences:
    for word in sentence.split():
```

```

        if word.lower() not in word_index:
            word_index[word.lower()] = index
            index += 1

# Convert text into sequences
X = []
for sentence in sentences:
    X.append([word_index[word.lower()] for word in sentence.split()])

X = tf.keras.preprocessing.sequence.pad_sequences(X, padding='post')
y = tf.keras.preprocessing.sequence.pad_sequences(labels, padding='post')

# Build GRU model
model = Sequential([
    Embedding(input_dim=50, output_dim=8),
    GRU(16, return_sequences=True),
    Dense(4, activation='softmax')
])

# Compile model
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy',
metrics=['accuracy'])

# Train model
model.fit(X, y, epochs=10)

# Predict entities
predictions = model.predict(X)
print(predictions)

```

Q2. English-to-Spanish Dictionary Program

```

# Dictionary to store translations
translations = {}

while True:
    print("\n1. Add Word")
    print("2. Search Word")
    print("3. Exit")

    choice = input("Enter choice: ")

    # Add new word and translation
    if choice == '1':
        english = input("Enter English word: ")
        spanish = input("Enter Spanish translation: ")
        translations[english] = spanish
        print("Word added successfully!")

```

```

# Search for translation
elif choice == '2':
    word = input("Enter English word: ")

    if word in translations:
        print("Spanish:", translations[word])
    else:
        print("Word not found")

# Exit program
elif choice == '3':
    break

else:
    print("Invalid choice")

```

Q3. Scatter Plot for Cluster Visualization

```

import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
import numpy as np

# Sample dataset
X = np.array([
    [1, 2], [1.5, 1.8], [5, 8],
    [8, 8], [1, 0.6], [9, 11]
])

# Apply K-Means clustering
kmeans = KMeans(n_clusters=2)
kmeans.fit(X)

# Cluster labels
labels = kmeans.labels_

# Scatter plot
plt.scatter(X[:, 0], X[:, 1], c=labels)
plt.title("Cluster Visualization")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()

```

Q4. Azure AI Service for Vision-Related Problem

Solution:

For a vision-related problem such as object detection, image classification, OCR, or facial recognition, the most suitable service is **Azure Computer Vision**.

Reason:

- Provides image analysis APIs
- Supports OCR for text extraction
- Detects objects and faces
- Easy integration with applications
- Cloud scalability and reliability

Use Cases:

- Security surveillance
 - Medical image analysis
 - Smart retail systems
 - Automated document scanning
-

Q5. Linear Regression Model for House Price Prediction

```
import pandas as pd
from sklearn.linear_model import LinearRegression

# Sample dataset
# Area in square feet and house prices

data = {
    'Area': [1000, 1500, 2000, 2500, 3000],
    'Price': [200000, 300000, 400000, 500000, 600000]
}

# Create DataFrame
_df = pd.DataFrame(data)

# Features and target
X = _df[['Area']]
y = _df['Price']

# Create linear regression model
model = LinearRegression()

# Train model
model.fit(X, y)

# Predict price for new area
```

```
predicted_price = model.predict([[2200]])  
  
print("Predicted Price:", predicted_price[0])
```

Lab 02

Q6. Basic RNN for Next Word Prediction

```
import numpy as np  
from tensorflow.keras.models import Sequential  
from tensorflow.keras.layers import Embedding, SimpleRNN, Dense  
from tensorflow.keras.preprocessing.text import Tokenizer  
from tensorflow.keras.preprocessing.sequence import pad_sequences  
  
# Training text  
text = "machine learning is fun and machine learning is powerful"  
  
# Tokenize text  
  
tokenizer = Tokenizer()  
tokenizer.fit_on_texts([text])  
  
# Create sequences  
input_sequences = []  
words = text.split()  
  
for i in range(1, len(words)):  
    sequence = words[:i+1]  
    encoded = tokenizer.texts_to_sequences([' '.join(sequence)])[0]  
    input_sequences.append(encoded)  
  
# Pad sequences  
max_len = max(len(seq) for seq in input_sequences)  
input_sequences = pad_sequences(input_sequences, maxlen=max_len,  
padding='pre')  
  
# Features and labels  
X = input_sequences[:, :-1]  
y = input_sequences[:, -1]  
  
# Build RNN model  
model = Sequential([  
    Embedding(50, 10, input_length=max_len-1),  
    SimpleRNN(50),  
    Dense(50, activation='softmax')  
])
```

```
# Compile model
model.compile(loss='sparse_categorical_crossentropy', optimizer='adam')

# Train model
model.fit(X, y, epochs=100)
```

Q7. Azure Metrics Advisor Usage

Solution:

Azure Metrics Advisor can be used to monitor business metrics automatically.

Steps:

1. Connect data source such as SQL database or Azure Data Explorer.
2. Configure metrics like sales, revenue, or website traffic.
3. Enable anomaly detection.
4. Use alerts to notify administrators when abnormal trends occur.
5. Use root cause analysis to diagnose issues.

Benefits:

- Real-time monitoring
- Automated anomaly detection
- Root cause analysis
- Better business decision-making

Q8. Azure Anomaly Detector Solution

Solution:

Azure Anomaly Detector can identify unusual behavior in time-series data.

Example Scenario:

Detecting abnormal spikes in online transactions.

Steps:

1. Collect time-series data.
2. Send data to Azure Anomaly Detector API.
3. Configure sensitivity level.
4. Analyze returned anomalies.
5. Trigger alerts for suspicious patterns.

Benefits:

- Detects fraud and failures

- Supports real-time monitoring
 - Uses machine learning algorithms
-

Q9. Managing Security for Azure AI Services

Solution:

Security can be improved using Azure Virtual Networks (VNETs).

Security Measures:

- Restrict public internet access.
- Use private endpoints.
- Apply network security groups.
- Enable encryption for data.
- Use Azure Active Directory authentication.

Benefits:

- Prevents unauthorized access
 - Secure communication between services
 - Better compliance and privacy
-

Q10. Azure Cognitive Service for Language Analysis

Scenario:

A company wants to analyze customer reviews to identify positive and negative feedback.

Service Used:

Azure Text Analytics

Reason:

- Performs sentiment analysis
- Extracts key phrases
- Detects language
- Recognizes named entities

Benefits:

- Improves customer experience
 - Automates feedback analysis
 - Saves time and effort
-

Q11. LSTM-Based Creative Text Generator

```
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences

# Sample text
text = "deep learning creates intelligent systems"

# Tokenization

tokenizer = Tokenizer()
tokenizer.fit_on_texts([text])

# Generate sequences
input_sequences = []
words = text.split()

for i in range(1, len(words)):
    seq = tokenizer.texts_to_sequences([' '.join(words[:i+1])])[0]
    input_sequences.append(seq)

# Padding
max_len = max(len(seq) for seq in input_sequences)
input_sequences = pad_sequences(input_sequences, maxlen=max_len,
padding='pre')

X = input_sequences[:, :-1]
y = input_sequences[:, -1]

# Build LSTM model
model = Sequential([
    Embedding(50, 10, input_length=max_len-1),
    LSTM(64),
    Dense(50, activation='softmax')
])

# Compile model
model.compile(loss='sparse_categorical_crossentropy', optimizer='adam')

# Train model
model.fit(X, y, epochs=100)
```

Lab 03

Q12. Azure Content Moderator Solution

Solution:

Azure Content Moderator helps detect inappropriate or offensive content.

Workflow:

1. User uploads text or images.
2. Data is sent to Content Moderator API.
3. Service checks for offensive language, violence, or adult content.
4. Unsafe content is flagged or removed.

Benefits:

- Improves platform safety
 - Reduces manual moderation effort
 - Supports text and image moderation
-

Q13. K-Means Clustering for Customer Segmentation

```
import pandas as pd
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Sample customer data

data = {
    'Annual_Spending': [200, 220, 250, 800, 850, 900],
    'Visits': [2, 3, 2, 10, 12, 11]
}

# Create DataFrame
_df = pd.DataFrame(data)

# Apply K-Means clustering
kmeans = KMeans(n_clusters=2)
_df['Cluster'] = kmeans.fit_predict(_df)

# Plot clusters
plt.scatter(_df['Annual_Spending'], _df['Visits'], c=_df['Cluster'])
plt.xlabel('Annual Spending')
plt.ylabel('Visits')
plt.title('Customer Clusters')
plt.show()
```

Q14. GRU-Based English to French Translator

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, GRU, Dense

# Simple GRU model
model = Sequential([
    Embedding(input_dim=1000, output_dim=64),
    GRU(128),
    Dense(1000, activation='softmax')
])

# Compile model
model.compile(optimizer='adam', loss='categorical_crossentropy')

print("GRU Translator Model Created Successfully")
```

Q15. Histogram of Exam Scores

```
import matplotlib.pyplot as plt

# Exam scores
scores = [55, 60, 65, 70, 75, 80, 85, 90, 95]

# Create histogram
plt.hist(scores, bins=5)

# Labels and title
plt.title("Distribution of Exam Scores")
plt.xlabel("Scores")
plt.ylabel("Frequency")

# Display plot
plt.show()
```

Q16. Correlation Coefficient Calculation

```
import pandas as pd

# Sample data
data = {
    'Hours_Studied': [1, 2, 3, 4, 5],
```

```

    'Marks': [20, 40, 50, 70, 90]
}

# Create DataFrame
_df = pd.DataFrame(data)

# Calculate correlation
correlation = _df['Hours_Studied'].corr(_df['Marks'])

print("Correlation Coefficient:", correlation)

```

Q17. LSTM Sentence Completion Model

```

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense

# Build LSTM model
model = Sequential([
    Embedding(input_dim=500, output_dim=64),
    LSTM(128),
    Dense(500, activation='softmax')
])

# Compile model
model.compile(optimizer='adam', loss='categorical_crossentropy')

print("Sentence completion model created")

```

Q18. Spam Email Classifier

```

from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.model_selection import train_test_split

# Sample emails
emails = [
    'Win money now',
    'Meeting at 5 PM',
    'Claim your free prize',
    'Project submission tomorrow'
]

# Labels
# 1 = Spam, 0 = Not Spam
labels = [1, 0, 1, 0]

```

```
# Convert text into numerical features
vectorizer = CountVectorizer()
X = vectorizer.fit_transform(emails)

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, labels, test_size=0.2)

# Train Naive Bayes model
model = MultinomialNB()
model.fit(X_train, y_train)

# Predict sample email
sample = vectorizer.transform(['Free money available'])
prediction = model.predict(sample)

if prediction[0] == 1:
    print('Spam Email')
else:
    print('Not Spam')
```